

RRL Abstracts — Sources Supporting VisioSphere

18 sources, all 2020 or later, all peer-reviewed. Abstract followed by exactly what it validates for this system.

Before you quote any of these

Abstracts were retrieved from publisher pages, the OpenAlex API or PubMed. Most are verbatim; a few were reconstructed from indexed data and read as structured summaries rather than the publisher's own wording. **Before quoting any abstract directly in your paper, open the link and copy from the publisher page.** Use these as a working reference for deciding which sources to read, not as quotable text. One entry — Blackburn *et al.* [10] — has no retrievable abstract and is marked accordingly.

Inclusion bar: peer-reviewed venue, 2020 or later, topically direct, and — for empirical papers — actual reported metrics. Three sources considered earlier were cut for failing it: a preprint that had not passed review, and two papers reporting no precision, recall or accuracy figures at all.

PART 1 — Sources supporting the detection directions

[1] W. Chen, Z. Jiang, H. Guo and X. Ni, "Fall detection based on key points of human-skeleton using OpenPose," *Symmetry*, vol. 12, no. 5, art. 744, 2020. <https://www.mdpi.com/2073-8994/12/5/744>

Falls are the primary cause of injury or death for the elderly over 65 years old. About 30% of the elderly over 65 years old fall every year. The reason for falling is that the center of gravity of the human body is not stable or symmetry breaking, and the body cannot keep balance. To solve the above problem, in this paper, we propose an approach for reorganization of accidental falls based on the symmetry principle. **We extract the skeleton information of the human body by OpenPose and identify the fall through three critical parameters: speed of descent at the center of the hip joint, the human body centerline angle with the ground, and width-to-height ratio of the human body external rectangular.** Unlike previous studies that have just investigated falling behavior, we consider the standing up of people after falls. **This method has 97% success rate to recognize the fall down behavior.**

VALIDATES The three-signal design at the core of VisioSphere — descent speed, trunk angle, bounding-box aspect ratio — used as three explicit decision conditions, with 97% reported accuracy. This is the closest published equivalent to your rule-based structure and your primary citation for the fall module.

[2] D. R. Beddiar, M. Oussalah and B. Nini, "Fall detection using body geometry and human pose estimation in video sequences," *J. Vis. Commun. Image Represent.*, vol. 82, art. 103407, 2021. <https://www.sciencedirect.com/science/article/pii/S1047320321002728>

According to the World Health Organization, falling is a significant health problem that causes thousands of deaths every year. Fall detection and fall prediction tasks enable accurate medical assistance to vulnerable populations whenever required. **We present in this paper a fall detection approach that explores human body geometry available at different frames of a video sequence. Especially, pose estimation, angle and distance between the vector formed by the head-centroid and the vector aligned with the horizontal axis of the hip body, and the center of hip, are employed to construct new distinctive image features.** A two-class Support Vector Machine (SVM) classifier and Temporal Convolution Network (TCN) are trained on newly constructed feature images. At the same time, a Long-Short-Term Memory (LSTM) network is trained on calculated angle and distance sequences to classify fall and non-fall activities. We perform experiments on Le2i FD dataset and UR FD dataset, where we also propose a cross-dataset evaluation. The results demonstrate the effectiveness and efficiency of the developed approach.

VALIDATES Deriving fall features from *body geometry computed on pose keypoints* across a video sequence — the same pipeline shape as yours. Also supports analysing geometry over a sequence rather than classifying isolated frames, which is what your 3-frame confirmation does.

[3] V. Kong, S. Soeng, M. Thon, W.-S. Cho, A. Nayyar and T.-K. Kim, "PIFR: A novel approach for analyzing pose angle-based human activity to automate fall detection in videos," *PLOS ONE*, vol. 20, no. 6, art. e0325253, 2025. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0325253>

Falls pose a significant health risk for elderly populations, necessitating advanced monitoring technologies. This study introduces a novel two-stage fall detection system that combines computer vision and machine learning to accurately identify fall events. **The system uses the YOLOv11 object detection model to track individuals and estimate their body pose by identifying 17 key body points across video frames. The proposed approach extracts nine critical geometric features, including the center of mass and various body angles.** These features are used to train a support vector machine (SVM) model for binary classification, distinguishing between standing and lying with high precision. **The system's temporal validation method analyzes sequential frame changes, ensuring robust fall detection.** Experimental results, evaluated on the University of Rzeszow Fall Detection (URFD) dataset and the Multiple Cameras Fall Dataset (MCFD), demonstrate exceptional performance, achieving **88.8% precision, 94.1% recall, an F1-score of 91.4%, and a specificity of 95.6%.**

VALIDATES YOLOv11 and 17 COCO keypoints — the same perception stack you run. Supports multi-angle geometric features, temporal validation across sequential frames, and gives you the exact metric set and benchmarks (URFD, MCFD) to report your own evaluation against.

[4] A. R. Inturi, V. M. Manikandan, M. N. Kumar, S. Wang and Y. Zhang, “Synergistic integration of skeletal kinematic features for vision-based fall detection,” *Sensors*, vol. 23, no. 14, art. 6283, 2023.

<https://www.mdpi.com/1424-8220/23/14/6283>

Each year, thousands of people die as a result of falls, with seniors making up 80% of these fatalities. Our study focuses on developing a vision-based fall detection system. **Our work proposes a new feature descriptor that results in a new fall detection framework. The body geometry of the subject is analyzed and patterns that help to distinguish falls from non-fall activities are identified.** An AlphaPose network is employed to identify 17 keypoints on the human skeleton. Thirteen keypoints are used in our study, and we compute two additional keypoints. **These 15 keypoints are divided into five segments... These five segments represent the left hand, right hand, left leg, right leg and craniocaudal section. A novel feature descriptor is generated by extracting the distances from the segmented parts, angles within the segmented parts and the angle of inclination for every segmented part... giving us 15 features per frame that preserve spatial information. To capture temporal dynamics, the extracted spatial features are arranged in the temporal sequence.** Our system was evaluated on the UPfall dataset, which is a benchmark dataset.

VALIDATES Supports your use of segment distances, joint angles and angles of inclination from 17 COCO keypoints. Also your best citation for future work: it shows the direction a richer feature set takes — 15 features per frame stacked temporally, rather than a handful thresholded per frame.

[5] V. Dentamaro, D. Impedovo and G. Pirlo, “Fall detection by human pose estimation and kinematic theory,” in *Proc. 25th Int. Conf. Pattern Recognition (ICPR)*, 2021, pp. 2328–2335. <https://ieeexplore.ieee.org/document/9413331>

In a society with increasing age, the understanding of human falls is of paramount importance. This paper presents a Decision Support System whose pipeline is designed to **extract and compute physical domain's features** achieving the state of the art accuracy on Le2i and UR fall detection datasets. The paper uses the **Kinematic Theory of Rapid Human Movement and its sigma-lognormal model together with classic physical features** in automatic fall detection on URFD datasets. The effort made in the design of this work toward recognition of falls uses **physical models whose laws are clear and understandable.**

VALIDATES Your closest in-window support for using *movement dynamics* rather than posture alone to identify a fall — the principle behind your 45 deg/s angular-rate boundary. The final sentence also supports your choice of interpretable physical models over a black-box classifier.

[6] A. Nagaj, Z. Li, D. P. Papadopoulos and K. Nasrollahi, “Visual context-aware person fall detection,” in *Smart Innovation, Systems and Technologies*, Springer, 2025, pp. 215–226. <https://arxiv.org/abs/2404.08088>

As the global population ages, the number of fall-related incidents is on the rise. Effective fall detection systems, specifically in the healthcare sector, are crucial to mitigate risks associated with such events. **This study evaluates the role of visual context, including background objects, on accuracy of fall classifiers.** We present a segmentation pipeline to semi-automatically separate individuals and objects in images. Well-established models like ResNet-18, EfficientNetV2-S, and Swin-Small were trained and evaluated. **Our findings highlight the significant influence of visual context on detection... Background objects such as beds, chairs, or wheelchairs can challenge fall detection, leading to false positive alarms. However, we demonstrate that object-specific contextual transformations during training effectively mitigate this challenge.** Further analysis using saliency maps supports our observation that visual context is crucial to classification tasks.

VALIDATES Direct support for your furniture zones. It states explicitly that beds, chairs and wheelchairs cause *false positive alarms* in fall detection and that handling them improves accuracy — which is the entire reason your `furniture_zones.json` polygons exist.

[7] Y. Zhang et al., “ByteTrack: Multi-object tracking by associating every detection box,” in *Computer Vision – ECCV 2022*, LNCS vol. 13682, Springer, 2022, pp. 1–21. <https://arxiv.org/abs/2110.06864>

Multi-object tracking (MOT) aims at estimating bounding boxes and identities of objects in videos. **Most methods obtain identities by associating detection boxes whose scores are higher than a threshold. The boxes with low scores, such as occluded objects, are simply thrown away, which brings non-negligible true object missing and fragmented trajectories.** To solve this problem, we present a simple, effective generic association method, **tracking almost every detection box instead of only the high score ones. For low score boxes, we utilize their similarities with tracklets to recover true objects and filter out the background detections.** When applied to 9 different state-of-the-art trackers, our method achieves consistent improvement on IDF1 ranging from 1 to 10 points... we achieve 80.3 MOTA, 77.3 IDF1 and 63.1 HOTA on the test set of MOT17 with 30 FPS running speed.

VALIDATES The tracker you actually run, and the direct justification for your tuning. The abstract names your exact problem: discarding low-score boxes causes missing objects and fragmented trajectories — which is why you lowered `track_high_thresh` to 0.15 and `new_track_thresh` to 0.10 so a body on the floor still receives a track ID.

[8] A. Zeng, L. Yang, X. Ju, J. Li, J. Wang and Q. Xu, “SmoothNet: A plug-and-play network for refining human poses in videos,” in *Computer Vision – ECCV 2022*, Springer, 2022, pp. 625–642. <https://arxiv.org/abs/2112.13715>

When analyzing human motion videos, the output jitters from existing pose estimators are highly-unbalanced with varied estimation errors across frames. Most frames in a video are relatively easy to estimate and only suffer from slight jitters. In contrast, for rarely seen or occluded actions, the estimated positions of multiple joints largely deviate from ground truth values for consecutive sequence of frames, rendering significant jitters on them. To tackle this problem, we propose to attach a dedicated temporal-only refinement network for jitter mitigation, named SmoothNet... SmoothNet learns the natural smoothness characteristics in body movements by learning long-range temporal relations of every joint. Comprehensive experiments on five datasets with eleven popular backbone networks for 2D and 3D pose estimation demonstrate the efficacy of the proposed solution.

VALIDATES Establishes that per-frame pose output is temporally unreliable, and that jitter is worst precisely for *occluded and rarely-seen poses* — which is exactly a person on the floor. Justifies both your keypoint EMA smoothing and your multi-frame confirmation requirement.

[9] F. L. Marshall, S. Zhang and B. Scotney, “Automatic assessment of the type and intensity of agitated hand movements,” *J. Healthcare Informatics Research*, vol. 6, no. 4, pp. 401–422, 2022.

<https://link.springer.com/article/10.1007/s41666-022-00120-3>

With increasing numbers of people living with dementia, there is growing interest in the automatic monitoring of agitation. Current assessments rely on carer observations within a framework of behavioural scales. **Automatic monitoring of agitation can supplement existing assessments, providing carers and clinicians with a greater understanding of the causes and extent of agitation. Despite agitation frequently manifesting in repetitive hand movements, the automatic assessment of repetitive hand movements remains a sparsely researched field.** Monitoring hand movements is problematic due to the subtle differences between different types of hand movements and variations in how they can be carried out; the lack of training data creates additional challenges. **This paper proposes a novel approach to assess the type and intensity of repetitive hand movements using skeletal model data derived from video...** Our results suggest that **data captured using a single conventional RGB video camera can be used to automatically monitor agitated hand movements of sedentary patients.**

VALIDATES The core citation for your agitation module: repetitive hand movement as a measurable agitation signal, from skeletal keypoints on a single RGB camera. It also states the field is *sparsely researched*, which supports your point about thin literature for this specific detection.

[10] J. Blackburn, K. Ousey, J. Stephenson and S. Lui, “Exploring the impact of experiencing a long lie fall on physical and clinical outcomes in older people requiring an ambulance: A systematic review,” *Int. Emergency Nursing*, vol. 62, art. 101148, 2022. (PMID 35245728)

<https://www.sciencedirect.com/science/article/abs/pii/S1755599X22000052>

Abstract not retrievable through open sources at the time of writing — the publisher record is paywalled and no abstract is indexed in OpenAlex. Verified record: systematic review examining prolonged time spent on the ground following falls among older people who required emergency ambulance services, and the resulting physical and clinical outcomes. **Retrieve the abstract from the publisher page or your library before citing.**

VALIDATES The clinical justification for detecting immobility at all, and for escalating rather than falling silent as duration grows. This is the source that supports your prolonged-fall ladder and the existence of your inactivity module — though not the specific 1200 s value.

[11] J. Kubitzka, M. Haas, L. Keppeler and B. Reuschenbach, “Therapy options for those affected by a long lie after a fall: A scoping review,” *BMC Geriatrics*, vol. 22, no. 1, art. 582, 2022.

<https://bmgeriatr.biomedcentral.com/articles/10.1186/s12877-022-03258-2>

BACKGROUND: Following a fall, **more than half of older people living alone are unable to get up or help independently.** This situation creates awareness of functional limitations and vulnerability. Patient-centered interventions addressing physical, psychological, and social needs are essential for managing this special population. **METHODS:** A six-step scoping framework aligned with modified PRISMA guidelines; literature searches across five databases and ten online archives; nineteen research papers met selection criteria. **RESULTS:** Two primary themes emerged — acute interventions managing the pronounced physical effects of prolonged recumbency, and preventive approaches examining standing training, assistive devices, and **monitoring systems** within fall prevention contexts. **CONCLUSIONS:** The evidence base remains limited regarding structured interventions for this population.

VALIDATES Establishes the “long lie” as a recognised clinical problem and names monitoring systems as one of the responses to it — placing VisioSphere’s inactivity and prolonged-fall modules inside an existing clinical framework rather than presenting them as a novelty.

[12] E. A. M. Michels, S. Gilbert, I. Koval and M. Wekenborg, “Alarm fatigue in healthcare: A scoping review of definitions, influencing factors, and mitigation strategies,” *BMC Nursing*, vol. 24, no. 1, art. 664, 2025.

<https://link.springer.com/article/10.1186/s12912-025-03369-2>

BACKGROUND: Alarm Fatigue is recognized as a significant risk to both patient safety and the well-being of healthcare professionals. However, it remains an underexplored phenomenon, further complicated by the lack of a harmonized definition. **METHODS:** A scoping review following PRISMA-ScR; systematic search in PubMed, CINAHL, MEDLINE and Google Scholar covering literature to April 2024. **RESULTS:** 32 publications were included. **Most described Alarm Fatigue as a phenomenon in which repeated exposure to frequent or non-actionable alarms leads to sensory overload, emotional strain, gradual desensitization, or reduced responsiveness among healthcare professionals, increasing risk of delayed or inadequate alarm responses and compromising patient safety. Identified mitigation strategies include training programs, technical improvements, protocol adjustments, and broader organizational interventions.**

VALIDATES The evidence that turns your escalation ladders and 30-alerts-per-hour circuit breaker from an engineering compromise into a patient-safety measure. Repeated non-actionable alarms cause desensitisation and delayed response; *technical mitigation* is a recognised strategy.

[13] E. Bayen *et al.*, “Reduction of time on the ground related to real-time video detection of falls in memory care facilities: Observational study,” *J. Medical Internet Research*, vol. 23, no. 6, art. e17551, 2021.

<https://pubmed.ncbi.nlm.nih.gov/34137723/>

BACKGROUND: Extended periods lying on the floor following a fall represent a critical factor affecting prognosis in elderly individuals. **Beyond immediate trauma-related injuries, prolonged immobilization leads to numerous comorbidities and potentially doubles mortality risk. OBJECTIVE:** Examined how SafelyYou Guardian technology — utilizing continuous video monitoring, artificial intelligence, and secure networks — affects early post-fall response. **METHODS:** Observational study over 10 months across 6 California memory care facilities, with a baseline phase, a weekly-review phase, and a real-time notification phase. **RESULTS:** Among 436 falls involving 66 participants (mean age 87), over 80% occurred in bedrooms with two-thirds happening overnight. **Real-time notification reduced time until assistance by 28.3 minutes and ground time by 29.6 minutes. The proportion of falls exceeding one-hour duration dropped from 31% at baseline to zero.**

VALIDATES The strongest single justification for the whole system. A deployed, client-based, per-facility AI camera system in memory care, evaluated in a peer-reviewed journal, showing that real-time notification cuts time on the ground by ~30 minutes. Cite it for why your prolonged-fall escalation matters clinically — and note that this system uses *staff video review*, not fully automatic alerting.

PART 2 — Sources validating “no single universal formula; the strongest recent research uses multiple complementary measurements”

The argument runs in three lines of evidence: **(A)** multiple recent systematic reviews survey the field and none identifies a single best method; **(B)** cross-domain benchmarking shows that no fixed feature set generalises between environments; **(C)** systems combining complementary measurements measurably outperform single-measurement baselines.

[14] M. Nabizade, R. Yahiaoui, I. Lajoie and N. Nacer, “Real-time vision-based fall detection systems for the elderly: A systematic review,” *Sensors*, vol. 26, no. 16, art. 5069, 2026. [Evidence line A]

<https://www.mdpi.com/1424-8220/26/16/5069>

Falls represent a threat to older adults, overload healthcare systems, and reduce quality of life. **Vision-based fall detection has advanced recently through deep learning, yet most proposed models lack validation on physical hardware and do not report inference-time metrics.** Following PRISMA and Kitchenham guidelines, this systematic review examined vision-based systems reporting inference speed on specified devices. Using a 10 fps threshold based on critical fall phase duration, the researchers **analyzed 588 records... identifying only 11 qualifying studies.** Key findings reveal CNN-based architectures dominate, edge devices constitute primary deployment platforms, and optimization techniques remain essential for real-time performance on constrained hardware. **The review identifies two significant limitations: absence of real-world testing with elderly participants and dependence on small datasets featuring simulated falls by younger volunteers.**

VALIDATES Two things at once. It shows the field is fragmented and under-validated — only 11 of 588 studies reported inference speed on real hardware — which supports your claim that no settled formula exists. It also makes your own work look *stronger* than most published systems: you have a real deployment in a real facility with a measured 12.18 fps on named hardware.

[15] A. Gattani, S. Dixit, M. Patil and M. Gupta, “Artificial intelligence for fall detection in older adults: A comprehensive survey of machine learning, deep learning approaches, and future directions,” *Ageing Research Reviews*, vol. 113, art. 102948, 2025. [Evidence lines A + C] <https://pubmed.ncbi.nlm.nih.gov/41265569/>

Fall detection systems are crucial for ensuring the safety of older adults... **However, developing accurate and reliable detection methods faces challenges due to the rarity of fall events and limited training data.** This review provides an in-depth examination of recent progress... This study follows the PRISMA 2020 guidelines to select articles on sensor- and vision-based methodologies and machine learning algorithms for fall detection. **Research findings indicate several strengths, weaknesses, and areas where accuracy can be improved...** This study introduces a taxonomy that categorizes fall detection methods according to the availability of data during classifier training. **Some major findings include the effectiveness achieved through sensor fusion and machine learning, aimed at improving accuracy in detecting falls, especially when sparse data are available.**

VALIDATES Supports both halves of your claim in one source. The need for a *taxonomy* of methods shows there is no single approach; and it names **fusion** — combining measurements — as a major route to improved accuracy. It also documents data scarcity, which explains why your own footage had to be the calibration source.

[16] F. X. Gaya-Morey, C. Manresa-Yee and J. M. Buades Rubio, “Deep learning for computer vision based activity recognition and fall detection of the elderly: A systematic review,” *Applied Intelligence*, vol. 54, no. 19, pp. 8982–9007, 2024. [Evidence line A] <https://link.springer.com/article/10.1007/s10489-024-05645-1>

As the proportion of elderly individuals in developed countries continues to rise globally, addressing their healthcare needs, particularly preserving their autonomy, is of paramount concern... This systematic review examines the literature pertaining to fall detection and Human Activity Recognition for elderly individuals... **A comprehensive search yielded 2,616 works from five distinct sources, spanning the years 2019 to 2023. From this pool, 151 relevant works were selected for detailed analysis. The review scrutinizes the employed DL models, datasets, hardware configurations, with particular emphasis on aspects such as privacy preservation and real-world deployment. The main contribution of this study lies in the synthesis of recent advancements... and identifying areas for further improvement.**

VALIDATES 151 distinct works analysed across five years, with the review's own contribution framed as synthesis and identification of gaps rather than endorsement of one method. If a universal formula existed, a review of this scope would name it.

[17] A. Raza, M. H. Yousaf, W. Ahmad, S. A. Velastín and S. Viriri, “Human fall detection using pose estimation: From traditional machine learning to vision transformers,” *Engineering Applications of Artificial Intelligence*, vol. 143, art. 109809, 2025. [Evidence line A] <https://www.sciencedirect.com/science/article/pii/S0952197624019687>

Human activity recognition research for healthcare has drawn global attention in recent era... Fall detection is crucial due to its potential fatality, especially for older individuals. Sensors are widely employed... **However, these devices may be uncomfortable or unsuitable for continuous use. Video imagery, in principle, surpasses sensors for fall detection.** The proposed method uses video frames to identify falls, reducing the need for sensors. **We present an empirical analysis of vision-based human fall detection, employing multiple techniques to estimate human poses including a transformer-based pose estimation technique. These techniques yield foundational features used for training diverse networks, including machine learning classifiers and vision transformers.** Our methodology achieves cutting-edge outcomes across the UR-Fall, UP-Fall, and Le2i fall detection datasets.

VALIDATES An empirical comparison of *multiple* pose techniques and *diverse* classifiers across three datasets — the structure of the paper is itself evidence that no single configuration is accepted as best. Also supports your choice of camera over wearables for continuous use in a care setting.

[18] H. Rehouma and M. Boukadoum, “Fall detection by deep learning-based bimodal movement and pose sensing with late fusion,” *Sensors*, vol. 25, no. 19, art. 6035, 2025. [Evidence line C] <https://www.mdpi.com/1424-8220/25/19/6035>

The timely detection of falls among the elderly remains challenging. **Single modality sensing approaches using inertial measurement units (IMUs) or vision-based monitoring systems frequently exhibit high false positives and compromised accuracy under suboptimal operating conditions.** We propose a novel bimodal deep learning-based sensing framework... by leveraging a memory-based autoencoder neural network for inertial abnormality detection and an attention-based neural network for visual pose assessment, with late fusion at the decision level. Our experimental evaluation... shows significant performance improvement by the described bimodal late-fusion system, with **an F1-score of 97.3% and, most notably, a false-positive rate of 3.6% significantly lower than the 11.3% and 8.9% with IMU-only and vision-only baselines, respectively.**

VALIDATES The quantitative proof for the “multiple complementary measurements” half of your claim. A single measurement source produced 8.9–11.3 % false positives; combining complementary measurements cut that to 3.6 %. This is the number to quote when you argue why your system uses two geometry paths plus several guards rather than one rule.

How to phrase the claim so it holds up

Phrase it from positive evidence rather than as an absence of evidence, and keep every clause tied to a peer-reviewed source in this list:

"No single measurement is sufficient for accurate fall, lying or agitation detection. Recent systems combine three to fifteen complementary geometric and kinematic features [1][3][4]; combining complementary measurement sources measurably outperforms single-source baselines, cutting false positives from 8.9–11.3 % to 3.6 % [18]; and systematic reviews spanning 151 to 588 studies identify no single best method, with most proposed models never validated on physical hardware or with elderly participants [14][16][17]."

What was lost by holding the strict bar: the sharpest evidence for the generalisation problem was a 2025 preprint (OmniFall) reporting accuracy collapsing from 100 % in-domain to 58.8 % on real-world footage. It is not peer-reviewed, so it is not in this list. Nabizade *et al.* [14] makes a similar point through review — absence of real-world testing with elderly participants, reliance on small datasets of simulated falls by younger volunteers — and is peer-reviewed. If your guidelines permit preprints, adding OmniFall back would strengthen the argument considerably.